

EFfiton: Design and Implementation of an Automated Integrated Reactor for Textile Wastewater Decolorization Using Homogeneous Electro-Fenton and Zeolite Adsorption with Real-Time Spectrophotometric Monitoring

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Abstract—Textile wastewater dye pollution is a persistent problem in Indonesia: measured dye concentrations of up to 1073.47 Pt-Co far exceed the 200 Pt-Co regulatory limit set by the Ministry of Environment and Forestry (Permen LHK No. 16/2019), requiring a degradation efficiency of at least 82% before treated effluent can be legally discharged. A prior UV-based adsorption-photo-Fenton system reached only 50% degradation while introducing UV-related safety, observability, and space constraints. This paper presents EFfiton, an automated benchtop reactor that integrates zeolite adsorption, homogeneous electro-Fenton (EF) degradation, and real-time diffraction-grating spectrophotometric monitoring into a single UV-free platform. A BJT-based voltage-controlled current source delivers linear electrode current across its full 0–220 mA design range ($R^2=1.000$), a FreeRTOS/ESP32 architecture coordinates 14 concurrent sensor and actuator tasks, and a custom TSL1401CL-based spectrophotometer (3.13 nm/pixel, 400–800 nm) tracks absorbance in real time. Operating the integrated system at 126 mA, methylene blue absorbance at $\lambda_{peak} = 721.3$ nm fell by 82% within one hour of EF treatment and by approximately 96% after two hours, meeting the system's core degradation target. Beyond degradation performance, the integrated system fully meets 9 of its 10 targeted product characteristics and all 18 of its detailed subsystem-level specifications, including a measured total input power of 192 W, well within its 250.4 W design budget. These results establish EFfiton as a functioning, automated, UV-free alternative capable of meeting Indonesia's textile wastewater discharge requirement.

Keywords—electro-Fenton, zeolite adsorption, textile wastewater treatment, methylene blue degradation, spectrophotometry, FreeRTOS, PID control, ESP32, automated reactor system, advanced oxidation process

I. INTRODUCTION

The Indonesian textile and clothing sector is one of the main drivers of the country's economy, contributing roughly IDR 218.2 trillion towards its national GDP in the year 2024, with annual growth rate of 4.26% [1]. This indicates that Indonesia's industrial production will increase significantly over time, especially in textile industries. However, the manufacturing process produces wastewater which contains synthetic dyes, including methylene blue, heavy metals, and complex organic compounds that are toxic, carcinogenic, and resistant to conventional biological treatment [2], [3]. Textile waste treatment, particularly chemical treatment, is not easy and requires specialized processing. Therefore, there is an urgent need to process these chemicals, which can be harmful to the environment. Regulatory standards established by the Indonesian Ministry of Environment and Forestry (Permen LHK No. 16/2019) state the maximum limit of color concentration in textile wastewater of 200 Pt-Co. However, actual industrial wastewater has been measured to contain a concentration of up to 1073.47 Pt-Co [4] which could be higher or lower, but we use this number as a reference. So, it means that the degradation of wastewater must be at least 82% before treated wastewater can be released legally into the environment.

Prior efforts made by BRIN and Sebelas Maret University (UNS) to develop an adsorption-photo-Fenton reactor filter only reached a degradation efficiency of 50% [5]. The results of this degradation efficiency can actually be further improved, considering that the electro-Fenton process itself is capable of degrading up to 89% of color [6]. On the other hand, this approach of photo-Fenton reactor relies on UV lamps which causes UV radiation that can be considered as an operational limitation regarding its observability, space

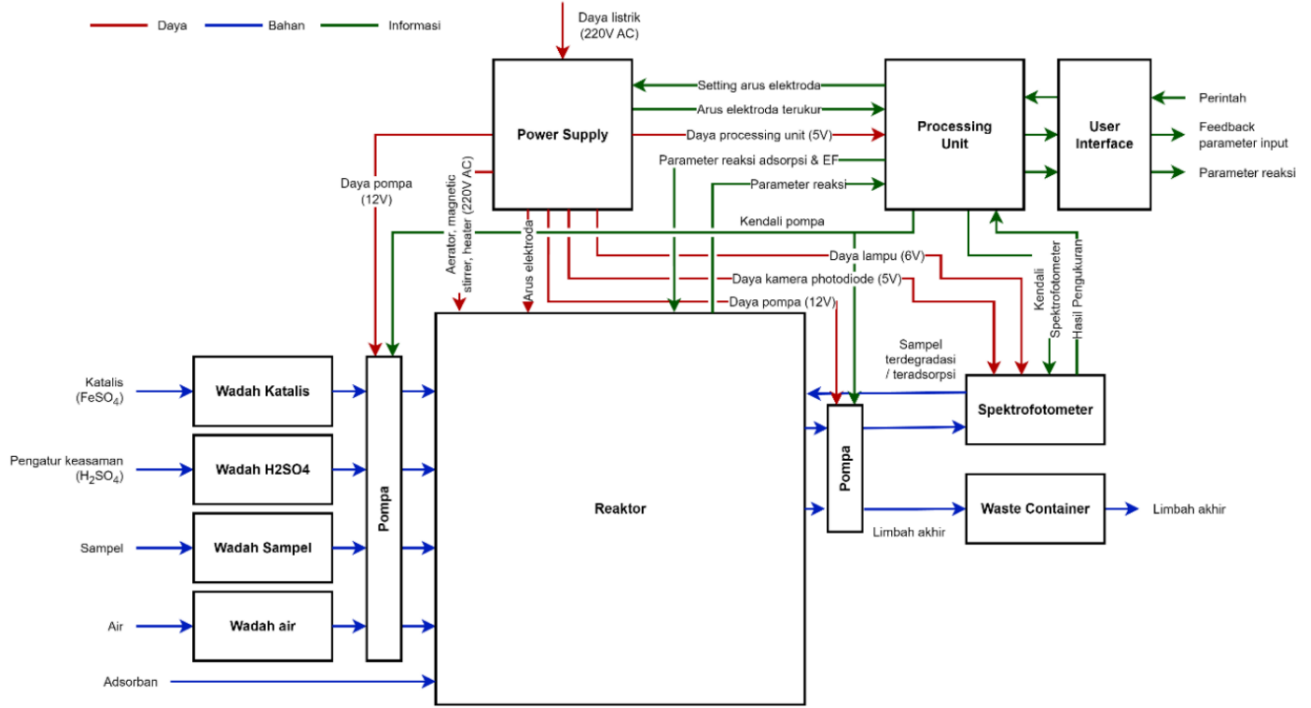
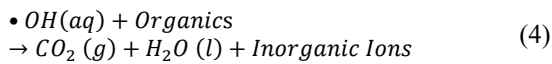
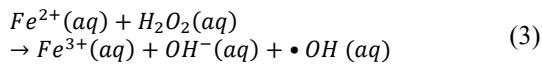
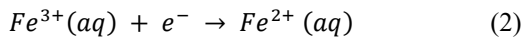
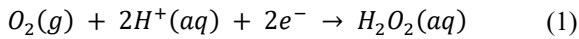


Fig. 1. Adsorption & electro-Fenton main system block diagram

constraints, and safety concerns. Therefore, the Fenton reaction using the electro-Fenton approach overcomes this limitation. Electro-Fenton is one of the Advanced Oxidation Process (AOPs) by producing H_2O_2 electrochemically through oxygen reduction at the cathode, while regenerating Fe^{2+} from Fe^{3+} entirely without UV and can be used again for upcoming reactions.



This study proposes Efiton, an automatic benchtop laboratory system designed to improve upon the previously developed UV-based solution. This system integrates zeolite adsorption, homogeneous Electro-Fenton degradation, and real-time spectrophotometer measurement into a single integrated platform. The contribution of this paper is therefore the integrated Efiton system as a whole, the coordinated operation of its adsorption, electro-Fenton, and monitoring subsystems toward a single degradation outcome, not any individual subsystem considered in isolation.

II. SYSTEM OVERVIEW

TABLE I. PRODUCT CHARACTERISTICS

No	Product Characteristics
1	Capable of performing dye adsorption using MOF or zeolite
2	Capable of performing the Fenton reaction with 82% degradation efficiency
3	Capable of adjusting the Fenton reaction parameters
4	Capable of measuring samples using a spectrophotometer
5	Capable of performing a flushing process at the end of the reaction
6	Capable of performing adsorption and Fenton reactions automatically and on a schedule
7	Affordable price
8	Easy to use for researchers with a Bachelor's degree background
9	Energy-efficient
10	Can be used directly on a laboratory bench

Fig. 1 shows the Efiton system architecture, organized around eight functional blocks: (1) the reactor core, a 250 mL glass beaker fitted with the electrode pair, heater, stirrer, and gas/pH/temperature sensors; (2) a multi-rail power supply (83 V, 12 V, 6 V, 5 V) that separately feeds the electrodes, actuators, and logic; (3) a 7-inch touchscreen and keypad interface for scheduling and monitoring; (4) seven peristaltic pumps that route the fluids for dosing, circulation, and flushing; (5) reagent and sample containers holding the

dye sample, FeSO_4 , and H_2SO_4 stock solutions; (6) a custom spectrophotometer that samples reactor fluid through the cuvette line and reports absorbance in real time; (7) an ESP32 running FreeRTOS, which coordinates 14 concurrent sensor and actuator tasks across a five-state process (Input Initialization, Initialization, Idle, Execution, and Flushing); and (8) a waste container that receives the reactor discharge after flushing. Detailed target-vs-achieved specifications for every subsystem are presented in TABLE II. (Section IV); subsystem-level design choices are discussed in Section III only where they affect degradation performance, with full derivations available in the individual subsystem reports [7], [8], [9].

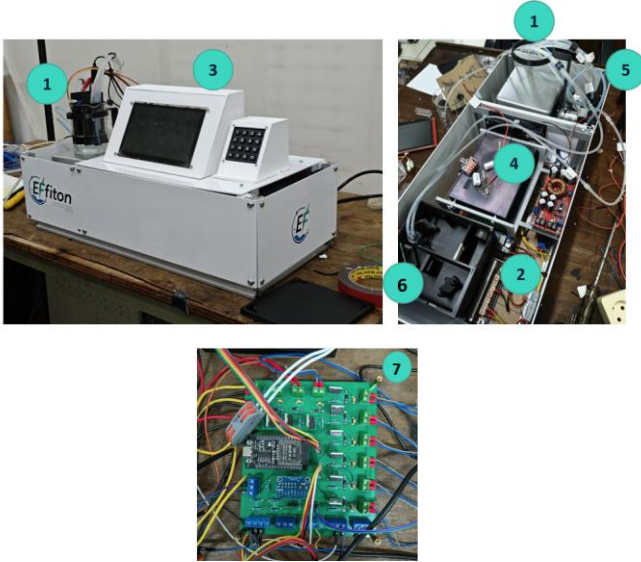


Fig. 2. Physical implementation of the EFFiton system: (1) electro-Fenton reactor, (2) power supply, (3) touchscreen and numpad user interface, (4) current source circuit, (5) pumps and reagent containers, (6) spectrophotometer, (7) Microcontroller PCB.

Based on the needs analysis we conducted to develop an automated benchtop laboratory system, we identified the characteristics of the product listed in TABLE I. and created a block diagram shown in Fig. 1, based on the results of our analysis. Generally speaking, these specifications reflect our original vision of a system integrated for zeolite adsorption, homogeneous Electro-Fenton degradation, and spectrophotometric measurement of degradation.

For the reactor, we conducted research on the Electro-Fenton reaction based on [6]. He and Zhou [10] and Kuleyin et al. [6] demonstrated how our Electro-Fenton reactor performed. The reactor blocks will work at pH 2-4 and oxygen aeration at 20 mL/min. The $\bullet\text{OH}$ radicals produced in (1) have an oxidation potential of 2.8 V against Standard Hydrogen Electrode (SHE) [10], which makes it one of the strongest oxidants known. This makes it highly effective in breaking down stubborn organic dye molecules [6]. An Electro-Fenton process is most efficient in an acidic environment (pH 2-4), where the production of H_2O_2 , and subsequently $\bullet\text{OH}$, is the highest [10]. Therefore, we adapted our equipment to achieve that environment so that the Fenton reaction could be carried out with an 82% degradation efficiency. Note also that the current source design meets the needs of the electro-Fenton system we will build.

The system will also include a user interface that will serve as a bridge between the user and our equipment. It will facilitate the automation of receiving commands to connect reagent/sample containers, seven peristaltic pumps, the reactor core, the spectrophotometer, and the waste container. Spectrophotometry must also be incorporated into the reaction scheduling for both the adsorption and electro-Fenton processes to track degradation changes every minute, thereby meeting our user need for the system to perform adsorption and Fenton reactions automatically and on a schedule.

III. METHODS AND IMPLEMENTATION

This section presents the reactor, dosing, current-source, spectrophotometric, and user-interface subsystems of EFFiton, focusing on how each design choice shapes the degradation outcome and product characteristics reported in Sections II and IV. Full derivations, PWM calibration curves, PID gain tuning, PCB routing, and mechanical drawings are presented in the individual subsystem reports [7], [8], [9].

A. Reactor System

The reactor integrates a magnetic stirrer, a closed-loop heater, and a monitoring sensor suite around the 250 mL electro-Fenton cell (Fig. 3). The magnetic stirrer facilitates the zeolite adsorption stage by keeping the adsorbent suspended and in contact with the dye solution, and supplies the reactor with both the kinetic energy of continuous mixing and the thermal energy of motor and fluid-shear heating that support the subsequent electro-Fenton stage. Because Fenton reaction rate is temperature-dependent, a closed-loop PID controller ($K_p = 6.977$, $K_i = 6.46 \times 10^{-3}$, $K_d = 1302.40$) regulates the reactor at 25-40 °C using a DS18B20 sensor and a solid-state-relay-switched heating element embedded in the stirrer base, coarse-heating to within 10 °C of setpoint before switching to PID control. Gas, pH, and temperature sensors are mounted directly on the reactor lid so the user can monitor reaction progress in real time from the touchscreen (Section III-E), directly supporting the product requirement that reaction status remain observable throughout adsorption and electro-Fenton operation.

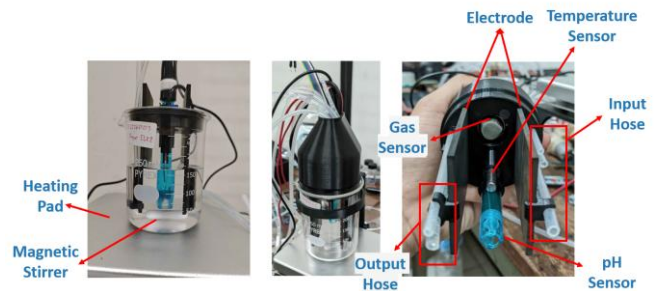


Fig. 3. Reactor core assembly, showing the heating pad and magnetic stirrer (left), assembled reactor housing (center), and electrode, temperature sensor, gas sensor, pH sensor, and input/output hose fittings (right).

B. Reagent Dosing

The FeSO_4 and H_2SO_4 concentrations set both the operating pH (2-4, required for efficient $\bullet\text{OH}$ generation) and the $\text{Fe}^{2+}/\text{H}_2\text{O}_2$ stoichiometry that drives the Fenton reaction. Two Kamoer NKP-DE-S10D peristaltic pumps deliver these reagents at a calibrated 30 mL/min,

individually PWM-tuned (170 and 187, respectively) to compensate for pump-to-pump flow variation; dosing volume is set by varying each pump's ON duration at this constant calibrated flow rate. Two additional pumps handle bulk sample transfer at 120 mL/min and drive the end-of-reaction flushing sequence with fresh water, directly satisfying the product's flushing requirement (TABLE I., item 5).

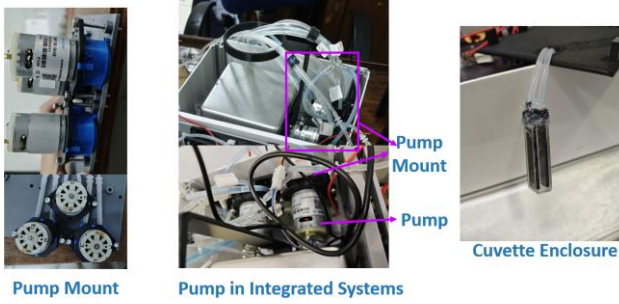


Fig. 4. Reagent dosing hardware, showing the pump mount (left), pump integrated in the fluidic system (center), and cuvette enclosure (right).

C. Electrode Current Source

The degradation rate is driven directly by the applied electrode current, which sets the rate of electrochemical H_2O_2 generation and $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$ regeneration ((1)–(2)). A BJT-based voltage-controlled current source (paralleled MJ15024 power transistors, LM358 feedback) converts an ESP32 DAC output to electrode current with the measured transfer function:

$$I_{\text{actual}} = 0.063 \cdot V_{\text{input}} + 0.00533 \text{ A} \quad (5)$$

This relationship gives $R^2 = 1.000$ across the full 0–220 mA design range, comfortably covering the 126 mA operating current used in Section IV; subsystem design and characterization are detailed in [7].

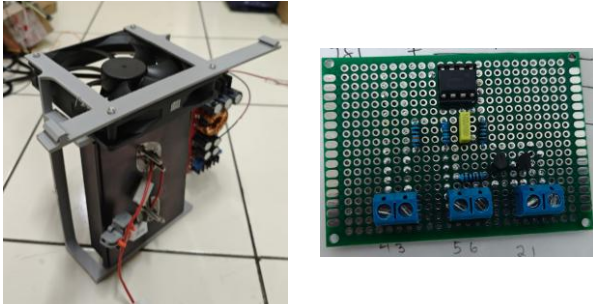


Fig. 5. BJT-based voltage-controlled current source circuit, mounted with its cooling fan, 83V PSU and enclosure (Left). Current Source Low Power Path as Control Circuit (Right)

D. Spectrophotometric Measurement

The degradation is quantified, not just observed, through a custom diffraction-grating spectrophotometer (600 lines/mm grating, TSL1401CL 128-pixel linear sensor, 3.13 nm/pixel resolution over 400–800 nm) that scans reactor fluid drawn through the cuvette line. Wavelength accuracy was verified against a laboratory spectrophotometer at PPNN ITB (1.67% error for methylene blue, 4.9% for a red dye reference), establishing confidence in the $\lambda_{\text{peak}} = 721.3$ nm degradation readings reported in Section TABLE IV.

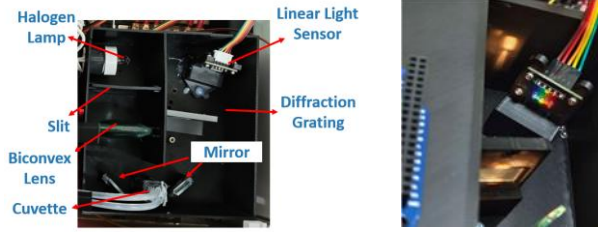


Fig. 6. Spectrophotometer internals, showing the halogen lamp, slit, biconvex lens, cuvette, mirror, diffraction grating, and linear light sensor.

E. User Interface and Microcontroller

In order to enhance usability, a 7-inch touchscreen with keypad fallback exposes scheduling, real-time sensor monitoring, and spectrophotometric visualization through a guided menu (Fig. 7), removing the need to interact directly with raw sensor data, firmware, or code. The same interface lets the user adjust reaction parameters, such as stirrer speed, temperature, electrode current, and reaction duration, and manually initiate the flushing sequence, directly supporting the product requirements for adjustable Fenton parameters and end-of-reaction flushing (TABLE I., items 3 and 5). Underlying this interface, an ESP32-V1 running FreeRTOS coordinates 14 independent sensor and actuator tasks in parallel through a five-state process (Input Initialization, Initialization, Idle, Execution, and Flushing), so that the scheduled adsorption-then-electro-Fenton sequence executes reliably and without race conditions; the user interface, firmware architecture, and supporting PCB integration are detailed in [7], [8].

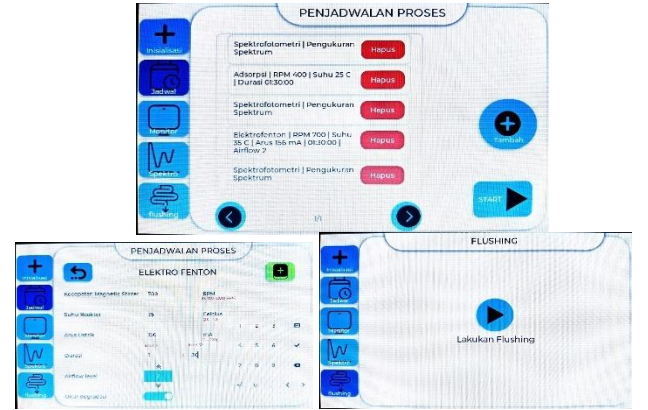


Fig. 7. Touchscreen interface, showing (top) the process scheduling menu, (bottom-left) reaction parameter adjustment, and (bottom-right) manual flushing control.

IV. RESULTS AND ANALYSIS

The central result of this work is the degradation performance of the fully integrated Effiton system. Fig. 8 shows the methylene blue sample vials collected at each measurement interval; Fig. 9 shows the corresponding absorbance spectrum (400–800 nm) recorded by the onboard spectrophotometer every 5 minutes over a 120-minute electro-Fenton run at 126 mA; and Fig. 10 tracks absorbance and percent degradation at $\lambda_{\text{peak}} = 721.3$ nm over the same run. Absorbance fell from 0.46 a.u. at $t = 0$ to 0.08 a.u. at $t = 60$ min, an 82.6% degradation efficiency, meeting the system's core specification (TABLE I., item 2) within the required one-hour timeframe, and continued to 0.02 a.u. ($\approx 96\%$ degradation) by $t = 120$ min. This exceeds the 50%

degradation reached by the prior UV-based adsorption-photo-Fenton system [5] while eliminating its UV-related safety, observability, and space constraints, and confirms that the electro-Fenton process is an effective system-level strategy rather than only a theoretical one. Degradation was measured at a single operating point (126 mA, within the full 0–220 mA design range established in Section III); mapping degradation rate against current density across the operating range is identified as future work



Fig. 8. Methylene blue sample vials collected throughout the 120-minute electro-Fenton run at 126 mA.

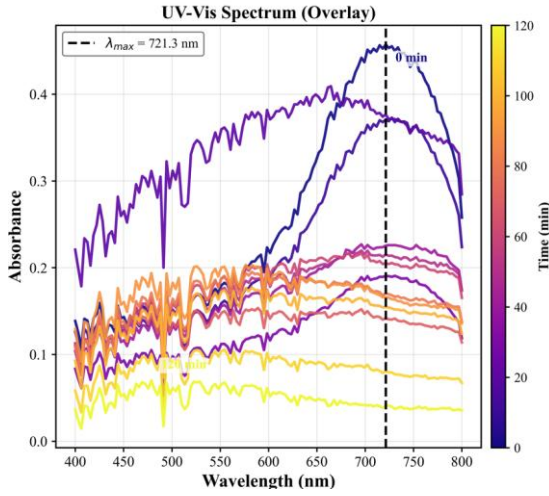


Fig. 9. UV-Vis absorbance spectra, 400–800 nm over a 120-minute electro-Fenton run at 126 mA.

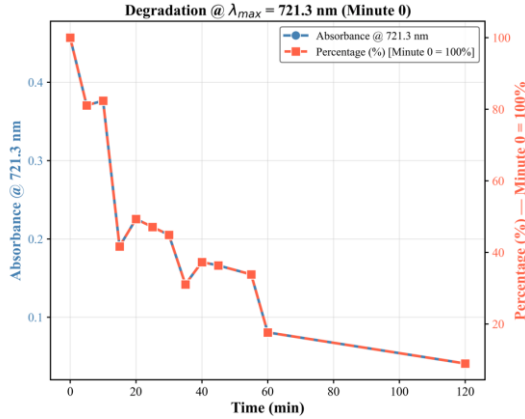


Fig. 10. Degradation curve at $\lambda_{peak} = 721.3$ nm over a 120-minute electro-Fenton run at 126 mA.

TABLE III. summarizes verification results for the subsystems that most directly support this outcome: the current source maintained $R^2 = 1.000$ linearity with negligible (0.003 V_{rms}) ripple; the PID-controlled heater held the reactor within its 25–40 °C target with a maximum capacity of 91.5 °C; the PI-controlled stirrer settled within ~4 s of a step setpoint change with zero steady-state error for speed above 100 rpm; and the dosing/transfer pumps met

their 30/120 mL/min targets after per-channel PWM calibration.

TABLE II. TARGET VS. ACHIEVED SPECIFICATION PER SUBSYSTEM

Subsystem	Target	Achieved
System input power	≤ 250.4 W	192 W measured (max. current, pumps, halogen lamp, and UI active)
Product dimensions	50×25×15 cm ³	50×25×15 cm ³
Touchscreen display	800×480 px	800×480 px
Spectrophotometer wavelength (measured)	400–800 nm	1.67% (MB) / 4.9% (red) error
Cuvette volume	3.5 mL	3.5 mL
Light source wavelength range	350–3500 nm	350–3500 nm
Light source power	20 ± 5% W	20 ± 5% W
Reactor electrode current	0–220 mA	0–220 mA ($R^2=1.000$)
Reactor pH	2–4	Calibrated, $R^2=0.97$
Magnetic stirrer speed	0–1200 rpm	100–1200 rpm stable
Reactor volume	250 mL	250 mL
Electrode submerged dimensions	5×3×0.3 cm	5×3×0.3 cm
Electrode spacing	3 cm	3 cm
Aerator flow rate	20 mL/min	20 mL/min
Transfer pump flow rate	120 mL/min	120 mL/min (PWM-calibrated)
Dosing pump flow rate	30 mL/min	30 mL/min (PWM-calibrated)
Reactor temperature	25–40 °C	25–40 °C (91.5 °C max. capacity)
Reaction duration	10–120 min	10–120 min

TABLE III. KEY SUBSYSTEM VERIFICATION RESULTS

Subsystem	Verification Result
Current source	$R^2=1.000$ linearity, 0.003 V _{rms} ripple
Heater / PID control	25–40 °C maintained, 91.5 °C max. capacity
Stirrer / PI control	~4 s settling, zero steady-state error (>100 rpm)
Dosing / transfer pumps	30 / 120 mL/min achieved after PWM calibration
Spectrophotometer	1.67% (MB) / 4.9% (red) wavelength error

Beyond the degradation figure itself, TABLE IV. evaluates the ten targeted product characteristics identified in TABLE I. . Nine of ten are fully met at the system level: dye adsorption, the 82% degradation target, adjustable Fenton parameters, spectrophotometric measurement, end-of-cycle flushing, scheduled/automatic operation, ease of use for a bachelor's-level researcher, energy efficiency (192 W measured total input power, within the 250.4 W design budget), and benchtop portability. Affordability was not yet evaluated quantitatively against a cost benchmark and is reported as pending. At the detailed subsystem-specification level (TABLE II.), all 18 items were met, and are documented alongside the other subsystem-level results in [7], [8], [9].

TABLE IV. PRODUCT CHARACTERISTICS ACHIEVEMENT SUMMARY

No	Product Characteristic	Status
1	Dye adsorption using zeolite	Met
2	82% Fenton degradation efficiency	Met (82.6% @ 126 mA, 60 min)
3	Adjustable Fenton reaction parameters	Met
4	Spectrophotometric sample measurement	Met
5	End-of-reaction flushing	Met

6	Automatic / scheduled operation	Met
7	Affordable price	Pending (not yet quantified)
8	Easy to use (bachelor's-level researcher)	Met
9	Energy-efficient	Met (192 W measured, within 250.4 W budget)
10	Usable directly on laboratory bench	Met

V. CONCLUSIONS

This work presents EFFiton, an automated benchtop system that integrates zeolite adsorption, homogeneous electro-Fenton degradation, and real-time spectrophotometric monitoring into a single UV-free platform for textile wastewater decolorization. Operated at 126 mA, the integrated system degraded methylene blue by 82.6% at $\lambda_{peak} = 721.3$ nm within one hour of electro-Fenton treatment, and by approximately 96% within two hours, meeting the system's core degradation requirement and outperforming the 50% achieved by the prior UV-based approach. At the system level, 9 of 10 targeted product characteristics and all 18 detailed subsystem-level specifications were met; only formal cost verification for affordability remains open. Future work includes: validating degradation performance of the fully integrated zeolite-adsorption-then-electro-Fenton sequence, which has not yet been tested as a combined process in this work (only the electro-Fenton stage was evaluated); stabilizing the spectrophotometer's mirror mount against vibration-induced spectral drift; increasing halogen lamp intensity and diffraction grating quality to reduce spectral noise and improve linearity calibration; improving low-RPM stirrer sensing (e.g., dual Hall sensors) to address motor stall; replacing the fluidic T-junction with a Y-junction to eliminate the observed 3.32% sample/water cross-contamination; and validating the system on real textile wastewater samples. Together, these results establish EFFiton as a working, automated, UV-free alternative for textile wastewater treatment that meets Indonesia's regulatory discharge requirement.

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